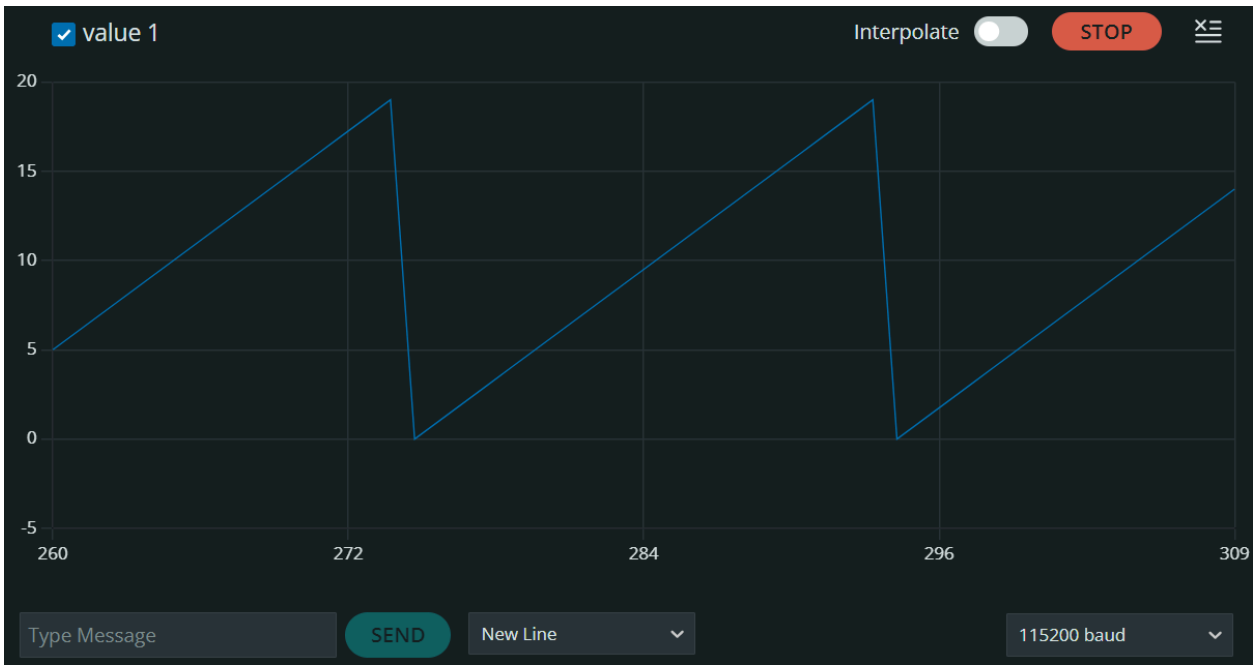
EE446-TinyML Lab 2

Author: Sparsh Dadhich

Github: https://github.com/sdadhich04/EE-446-TInyML-

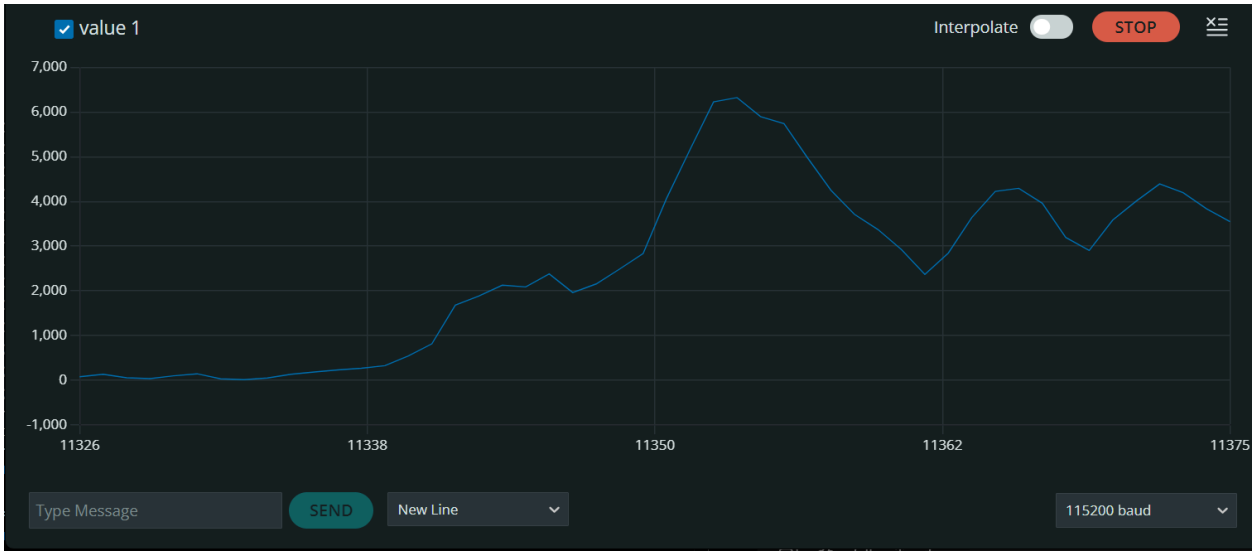
Task 4 :

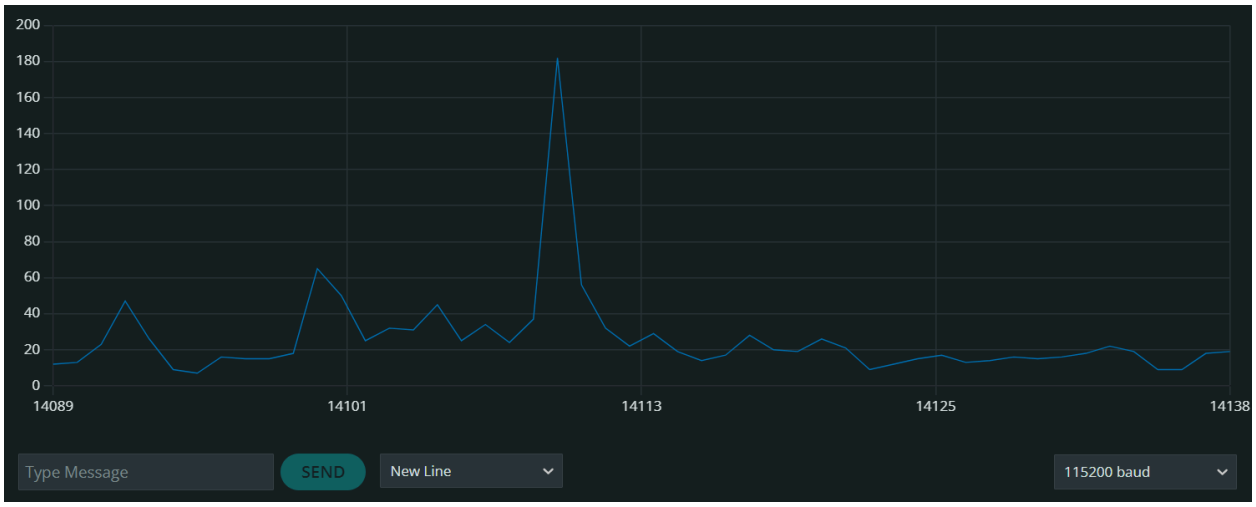
```
Serial test started
Count: 0
Count: 1
Count: 2
Count: 3
Count: 4
Count: 5
Count: 6
Count: 7
```



Task 5 : The cleanest change was created when I softly clapped near the MCU. It is the bottom most plot which shows a clear distinct sign of the event happening.







Task 6A : It is hard to say which orientation produced the biggest change but if I were to say which was the cleanest one it would be when i flipped the board upside down

```
-0.008,0.047,1.007
-0.007,0.047,1.009
-0.007,0.045,1.004
-0.007,0.045,1.008
-0.003,0.045,1.005
-0.006,0.046,1.008
-0.004,0.047,1.009
-0.009,0.046,1.007
-0.005,0.050,1.007
-0.007,0.048,1.009
-0.007,0.047,1.007
-0.005,0.046,1.007
-0.006,0.048,1.006
-0.009,0.046,1.007
-0.003,0.049,1.007
-0.005,0.046,1.005
-0.007,0.045,1.006
-0.005,0.044,1.007
-0.006,0.047,1.007
-0.007,0.048,1.009
-0.003,0.047,1.009
-0.015,0.047,1.011
-0.009,0.045,1.007
-0.006,0.046,1.003
-0.006,0.047,1.011
-0.008,0.047,1.007
-0.008,0.048,1.008
```

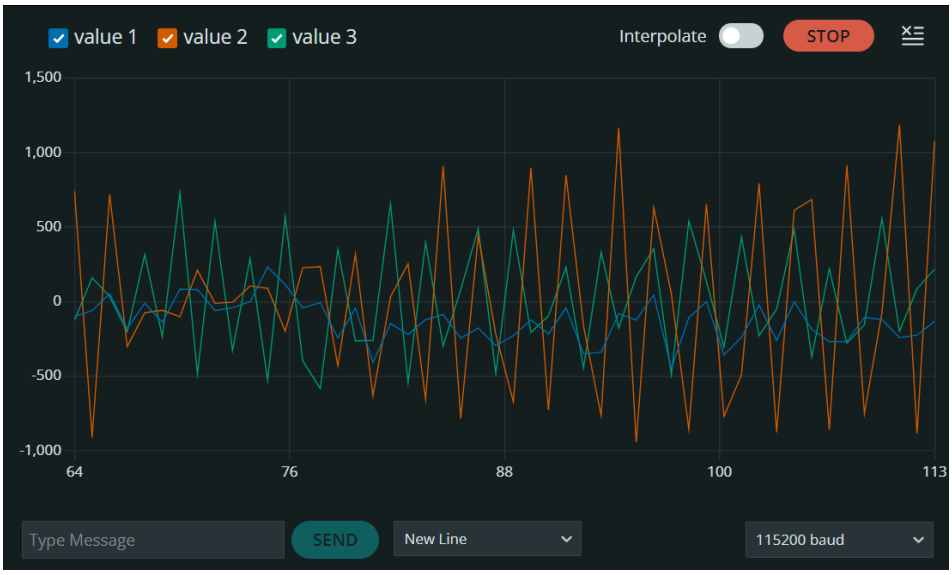
```
0.981,-0.299,-0.057
0.953,-0.232,0.077
0.897,-0.124,0.061
0.907,-0.180,0.060
1.015,-0.183,0.233
0.952,-0.200,0.057
0.919,-0.159,0.099
0.945,-0.134,0.156
0.966,-0.206,0.157
0.953,-0.205,0.138
0.941,-0.185,0.138
0.957,-0.191,0.162
0.963,-0.219,0.137
0.947,-0.179,0.147
0.960,-0.190,0.158
0.950,-0.150,0.156
0.960,-0.200,0.135
0.956,-0.141,0.178
0.961,-0.180,0.137
0.954,-0.167,0.148
0.967,-0.199,0.137
0.955,-0.189,0.137
0.968,-0.204,0.148
0.962,-0.188,0.154
0.962,-0.211,0.131
```

```
-1.001,0.009,0.110
-1.006,-0.003,0.075
-1.006,0.020,0.071
-1.005,0.010,0.059
-1.009,0.004,0.070
-1.009,0.006,0.042
-1.005,0.009,0.065
-1.004,0.006,0.070
-1.004,0.005,0.055
-1.006,0.016,0.060
-1.002,0.016,0.059
-1.006,0.003,0.041
-1.007,-0.000,0.044
-1.004,0.009,0.056
-1.003,0.013,0.062
-1.002,0.025,0.077
-1.006,0.017,0.069
-1.006,0.007,0.061
-1.006,0.013,0.062
-1.002,0.022,0.080
-1.004,0.019,0.073
-1.002,0.017,0.063
-1.008,0.018,0.043
```



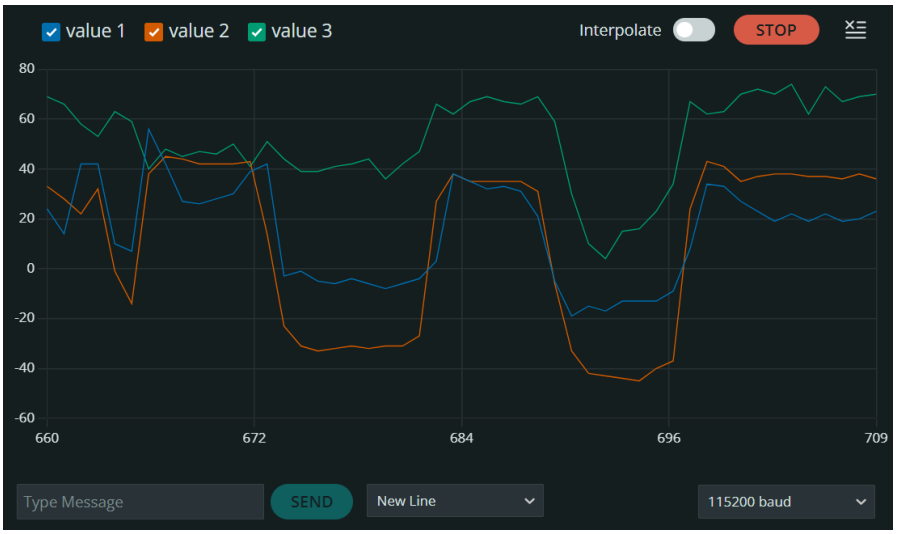
Task 6B: The cleanest change was when I rotated the board along its vertical axis.

```
1.160,0.061,0.000
0.793,0.000,0.000
-0.061,0.427,-0.366
0.793,0.000,0.061
2.625,-0.061,0.488
3.296,-0.366,0.854
1.953,0.183,0.366
2.502,-0.305,0.549
0.854,-0.122,-0.122
0.793,0.122,0.244
-0.549,0.061,-0.488
1.587,-0.183,0.305
0.671,-0.122,-0.244
-1.160,0.183,-0.732
0.122,0.366,-0.244
0.732,0.000,-0.061
2.319,0.000,0.488
0.854,0.000,0.000
-1.099,0.366,-0.610
-2.258,0.610,-0.916
-0.427,-0.061,-0.671
-0.183,0.366,0.061
2.930,-0.366,0.671
0.732,0.183,0.061
-0.854,0.427,-0.305
-0.305,-0.671,-1.099
1.953,-0.366,0.244
1.831,-0.061,0.793
1.343,0.183,0.366
0.305,0.244,0.061
-0.061,0
```



Task 6C : The cleanest change in the magnetometer readings was when I bought it close to my laptop.

```
45.000,-23.000,16.000
44.000,-26.000,17.000
45.000,-23.000,16.000
42.000,-23.000,22.000
34.000,-24.000,23.000
32.000,-21.000,23.000
32.000,-21.000,29.000
33.000,-22.000,25.000
34.000,-20.000,24.000
35.000,-21.000,29.000
34.000,-22.000,24.000
```



Task 7 : The cleanest change was noticed when I blew hot air onto the board which produced a huge spike in the Humidity levels.

```
Temperature (C): 25.64  Humidity (%): 26.13
Temperature (C): 25.64  Humidity (%): 26.13
Temperature (C): 25.62  Humidity (%): 26.13
Temperature (C): 25.62  Humidity (%): 26.15
Temperature (C): 25.61  Humidity (%): 26.15
Temperature (C): 25.61  Humidity (%): 26.16
Temperature (C): 25.61  Humidity (%): 26.19
Temperature (C): 25.60  Humidity (%): 26.21
Temperature (C): 25.56  Humidity (%): 26.23
Temperature (C): 25.56  Humidity (%): 26.21
Temperature (C): 25.55  Humidity (%): 26.19
Temperature (C): 25.53  Humidity (%): 26.13
Temperature (C): 25.53  Humidity (%): 26.10
Temperature (C): 25.53  Humidity (%): 26.06
```

```
Temperature (C): 27.53  Humidity (%): 43.05
Temperature (C): 26.44  Humidity (%): 58.82
Temperature (C): 27.47  Humidity (%): 67.45
Temperature (C): 28.33  Humidity (%): 73.35
Temperature (C): 27.85  Humidity (%): 75.90
Temperature (C): 28.84  Humidity (%): 79.37
Temperature (C): 28.64  Humidity (%): 80.62
Temperature (C): 28.68  Humidity (%): 80.91
Temperature (C): 28.87  Humidity (%): 81.06
Temperature (C): 28.95  Humidity (%): 82.09
```

Task 8 : The cleanest change in the pressure value was when I measured it at the top floor of Accolade apartments where the pressure value decreased.

```
Pressure (kPa): 101.662  Temperature (C): 26.63
Pressure (kPa): 101.668  Temperature (C): 26.63
Pressure (kPa): 101.664  Temperature (C): 26.63
Pressure (kPa): 101.662  Temperature (C): 26.63
Pressure (kPa): 101.660  Temperature (C): 26.62
Pressure (kPa): 101.665  Temperature (C): 26.61
Pressure (kPa): 101.665  Temperature (C): 26.61
Pressure (kPa): 101.668  Temperature (C): 26.61
Pressure (kPa): 101.664  Temperature (C): 26.60
Pressure (kPa): 101.659  Temperature (C): 26.60
Pressure (kPa): 101.667  Temperature (C): 26.61
```

Task 9A : The values change rapidly as I move my hand closer to the board and it is more noticeable from top down.

```
Proximity: 240
Proximity: 238
Proximity: 239
Proximity: 239
Proximity: 240
Proximity: 239
Proximity: 238
Proximity: 237
Proximity: 238
Proximity: 239
Proximity: 240
Proximity: 240
Proximity: 238
Proximity: 237
Proximity: 238
Proximity: 237
Proximity: 238
```

```
Proximity: 0
Proximity: 0
Proximity: 0
Proximity: 0
Proximity: 0
Proximity: 0
Proximity: 0
Proximity: 0
Proximity: 0
Proximity: 0
Proximity: 0
Proximity: 0
Proximity: 0
Proximity: 0
```

Task 9B: The right gesture was more easily recognised by the MCU.

```
Gesture test started
RIGHT
Gesture test started
LEFT
```

Task 9C : Flashing the light directly to the board produced the most amount of change in the system.

```
73,44,39,138
79,48,43,151
75,44,40,140
76,46,41,144
78,47,42,148
73,44,39,139
79,47,43,151
75,44,40,141
76,46,41,144
78,47,42,148
73,44,39,139
79,48,43,151
74,44,39,140
76,46,41,144
78,47,42,148
75,45,41,143
79,47,43,151
73,43,39,138
```

```
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
4097,4097,4097,4097
```

Task 10: I initially started with very high Thresholds which made me unable to recreate most of the settings and then slowly started adjusting in accordance to the plots and serial readings that I had previously collected and then also based on the raw observations produced during the task.

```
raw,mic=17,clear=62,motion=0.010,prox=240
flags,sound=0,dark=1,moving=0,near=1
state,QUIET_BRIGHT_STEADY_FAR
raw,mic=19,clear=65,motion=0.010,prox=239
flags,sound=0,dark=1,moving=0,near=1
state,QUIET_BRIGHT_STEADY_FAR
raw,mic=6,clear=65,motion=0.010,prox=239
flags,sound=0,dark=1,moving=0,near=1
state,QUIET_BRIGHT_STEADY_FAR
raw,mic=13,clear=61,motion=0.009,prox=242
flags,sound=0,dark=1,moving=0,near=1
state,QUIET_BRIGHT_STEADY_FAR
raw,mic=6,clear=61,motion=0.008,prox=242
flags,sound=0,dark=1,moving=0,near=1
state,QUIET_BRIGHT_STEADY_FAR
```

```
raw,mic=41,clear=1,motion=0.020,prox=0
flags,sound=0,dark=0,moving=0,near=0
state,QUIET_DARK_STEADY_NEAR
raw,mic=68,clear=1,motion=0.023,prox=0
flags,sound=0,dark=0,moving=0,near=0
state,QUIET_DARK_STEADY_NEAR
raw,mic=13,clear=1,motion=0.023,prox=0
flags,sound=0,dark=0,moving=0,near=0
state,QUIET_DARK_STEADY_NEAR
raw,mic=7,clear=1,motion=0.025,prox=0
flags,sound=0,dark=0,moving=0,near=0
state,QUIET_DARK_STEADY_NEAR
```

```
raw,mic=98,clear=55,motion=0.011,prox=239
flags,sound=1,dark=1,moving=0,near=1
state,NOISY_BRIGHT_STEADY_FAR
```

```
raw,mic=55,clear=108,motion=0.124,prox=183
flags,sound=1,dark=1,moving=1,near=0
state,NOISY_BRIGHT_MOVING_NEAR
```

Task 11: The thresholds were decided based on noticing nominal readings and then based on them and introduction of the different conditions noticing the changes in the system and then changing them to make it so that the conditions are met. As for the cooldown time it was selected by checking the previous tasks and seeing what was the minimal and optimal time that should work.

```
raw,rh=44.50,temp=25.49,mag=151.86,r=28,g=19,b=13,clear=52
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
raw,rh=44.32,temp=25.48,mag=151.54,r=28,g=20,b=13,clear=53
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
raw,rh=44.17,temp=25.48,mag=148.68,r=28,g=20,b=13,clear=53
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
raw,rh=44.09,temp=25.46,mag=153.43,r=32,g=22,b=15,clear=59
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
raw,rh=44.05,temp=25.48,mag=151.61,r=32,g=22,b=15,clear=59
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BASELINE_NORMAL
```

```
raw,rh=44.38,temp=26.40,mag=119.77,r=37,g=23,b=21,clear=81
flags,humid_jump=0,temp_rise=0,mag_shift=1,light_or_color_change=1
FINAL_LABEL=MAGNETIC_DISTURBANCE_EVENT
raw,rh=44.63,temp=26.40,mag=116.42,r=37,g=23,b=21,clear=81
flags,humid_jump=0,temp_rise=0,mag_shift=1,light_or_color_change=1
FINAL_LABEL=MAGNETIC_DISTURBANCE_EVENT
raw,rh=44.45,temp=26.40,mag=116.50,r=38,g=24,b=21,clear=81
flags,humid_jump=0,temp_rise=0,mag_shift=1,light_or_color_change=1
FINAL_LABEL=MAGNETIC_DISTURBANCE_EVENT
raw,rh=44.23,temp=26.40,mag=115.37,r=38,g=24,b=21,clear=81
flags,humid_jump=0,temp_rise=0,mag_shift=1,light_or_color_change=1
FINAL_LABEL=MAGNETIC_DISTURBANCE_EVENT
```

```
raw,rh=67.80,temp=25.57,mag=150.02,r=33,g=22,b=15,clear=55
flags,humid_jump=1,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BREATH_OR_WARM_AIR_EVENT
raw,rh=66.81,temp=25.52,mag=147.03,r=32,g=25,b=17,clear=61
flags,humid_jump=1,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BREATH_OR_WARM_AIR_EVENT
raw,rh=64.74,temp=25.52,mag=150.71,r=32,g=25,b=17,clear=61
flags,humid_jump=1,temp_rise=0,mag_shift=0,light_or_color_change=0
FINAL_LABEL=BREATH_OR_WARM_AIR_EVENT
```

```
raw,rh=42.21,temp=26.26,mag=137.72,r=4097,g=4097,b=4097,clear=4097
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=1
FINAL_LABEL=LIGHT_OR_COLOR_CHANGE_EVENT
raw,rh=42.28,temp=26.27,mag=138.18,r=4097,g=4097,b=4097,clear=4097
flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=1
FINAL_LABEL=LIGHT_OR_COLOR_CHANGE_EVENT
```